

# qsim: Trotterized Open-System Simulation

## Exact Kraus Convergence

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Overview: demonstration that exact Kraus extraction removes circuit-level  $O(dt)$  bias and yields machine-precision per-step maps.

# Agenda

1. Problem and goals
2. Method: Strang splitting + ancilla dilation
3. Debugging summary and fixes
4. Exact Kraus extraction results
5. Convergence plots and Richardson extrapolation
6. Conclusions and next steps

# Problem & Method

Goal: simulate Lindblad dynamics for 2 qubits using Trotterized circuits with ancilla-based amplitude damping dilation.

Method highlights:

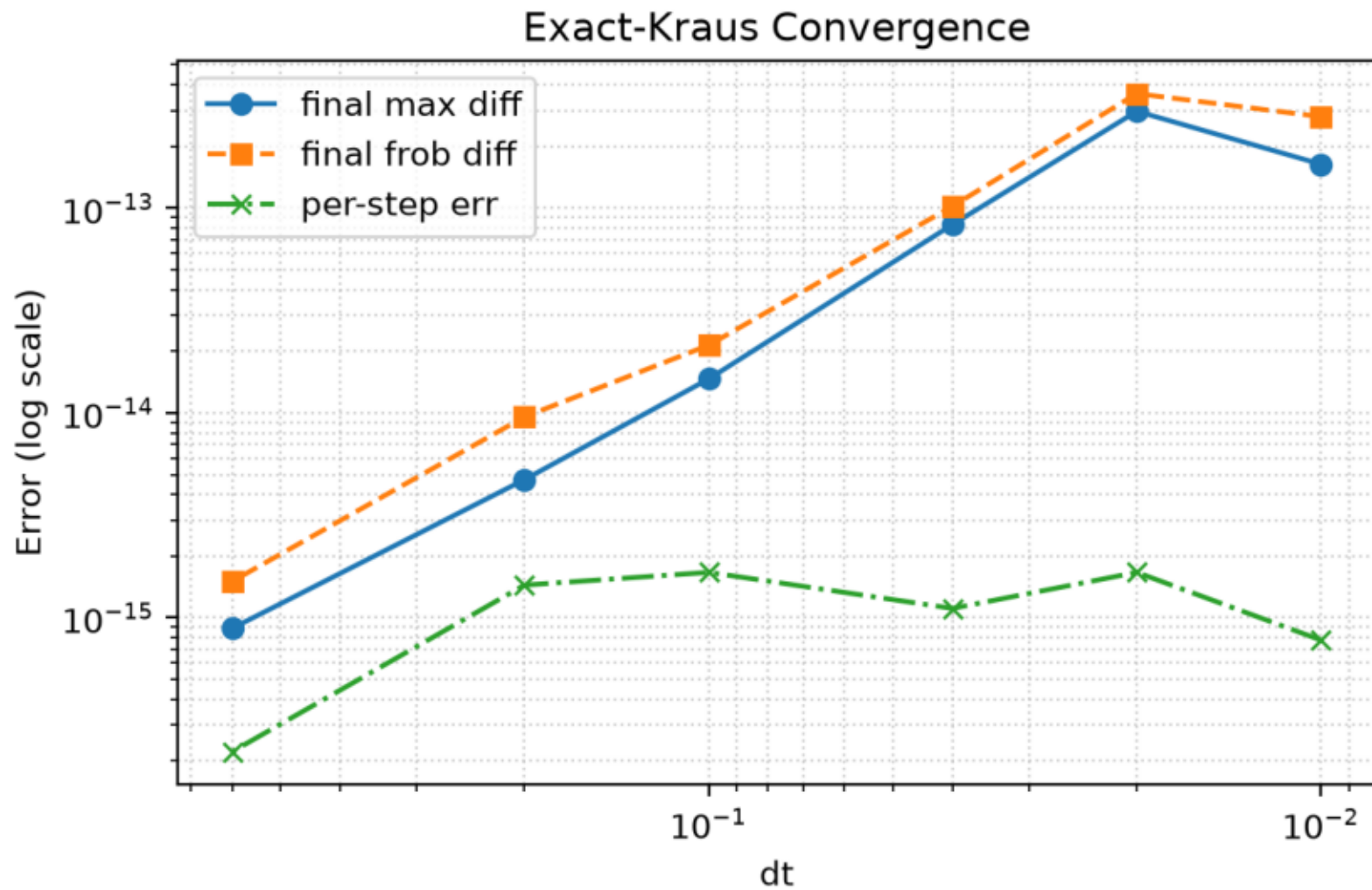
- Strang symmetric splitting:  $\exp(L_D dt/2) \exp(L_H dt) \exp(L_D dt/2)$
- Ancilla dilation for amplitude damping per qubit
- Kraus extraction from Choi root to obtain exact per-step maps

Validation: compare circuit Trotter outputs to classical QuTiP ground truth.

# Debugging summary (key fixes)

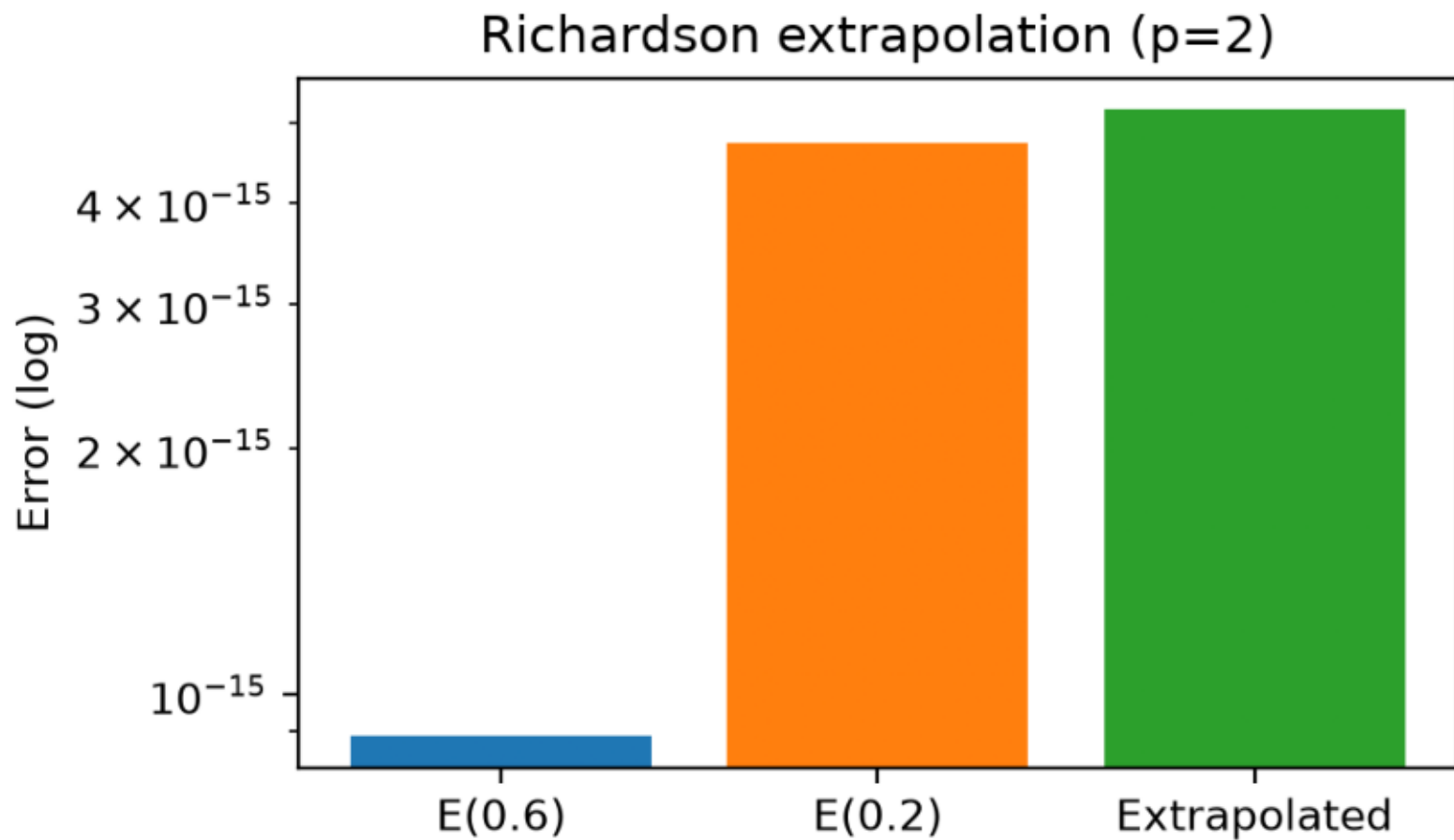
- Fixed conditional swap embedding; use unconditional swap for consistency.
- Use exact finite-time parameter:  $\text{gamma\_step} = 1 - \exp(-\text{gamma} \cdot \text{dt})$ .
- Resolved Choi  $\leftrightarrow$  Kraus reshape convention via transpose before normalization.
- Enforced Kraus completeness with polar/sqrt normalization to get  $S_{\text{from\_K}} \approx P$  to  $1e-15$ .

# Convergence: Errors vs dt (log-log)



Per-step error at machine precision; final errors  $\sim 1e-15..1e-13$

## Richardson extrapolation (p=2)



Extrapolated error and original errors (log scale)

# Report excerpt

# Conclusions & Next steps

- Exact-Kraus approach removes structural  $O(dt)$  per-step bias (per-step errors  $\sim 1e-15$ ).
- Final-state differences are at floating-point noise levels; method is validated.
- Next: (recommended) Richardson extrapolation on observables to produce publication-ready estimates; (optional) synthesize exact Kraus into implementable circuits.
- I can prepare slides as PDF (done) and a short markdown write-up if desired.